Indian Institute of Technology (BHU), Varanasi

Optimising Fed-Batch Bioprocesses using pyFOOMB

UG Project Progress Report

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Lab: BPDD (_____)

Project Overview

Objective

Optimise the **fed-batch feeding strategy** for recombinant protein production by exploiting the relationship between specific growth rate (μ) and specific product formation rate (q_p).

This Semester — Completed:

- ✓ Studied the pyFOOMB framework
- ✓ Built a full web-based GUI
- ✓ Surveyed μ–q_p models from literature
- ✓ Simulated batch X, S, P profiles

Next Steps:

- Implement fed-batch models in pyFOOMB
- Parameter estimation from real data
- Feed profile optimisation
- Validate with experimental data

What is pyFOOMB?

Python **F**ramework for **O**bject-**O**riented **M**odelling of **B**ioprocesses

Hemmerich et al. (2021), *Eng. Life Sci.* 21(3-4):242–257

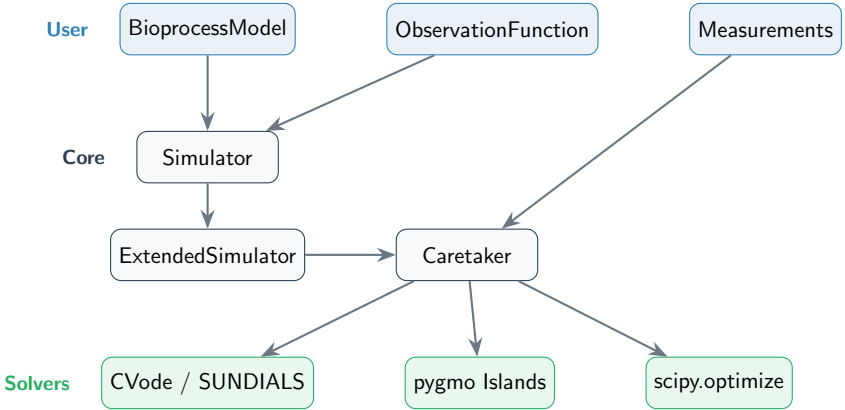
Core Capabilities:

- ODE-based bioprocess modelling
- User-defined kinetics in `rhs()` method
- Event handling for fed-batch (feed start/stop, bolus)
- Forward simulation via CVode (SUNDIALS)
- Parameter estimation — local (scipy) & global (pygmo)
- Sensitivity analysis, FIM, OED

Key Difference:

pyFOOMB does **not** hardcode kinetic models. Users implement *any* μ–q_p relationship in their model's `rhs()` method. The framework handles simulation, events, and estimation.

pyFOOMB Architecture



Users define ODE models → pyFOOMB handles integration, estimation & analysis

Web GUI — What We Built

Problem: pyFOOMB is code-only (Jupyter notebooks). No accessible interface for non-programmers.

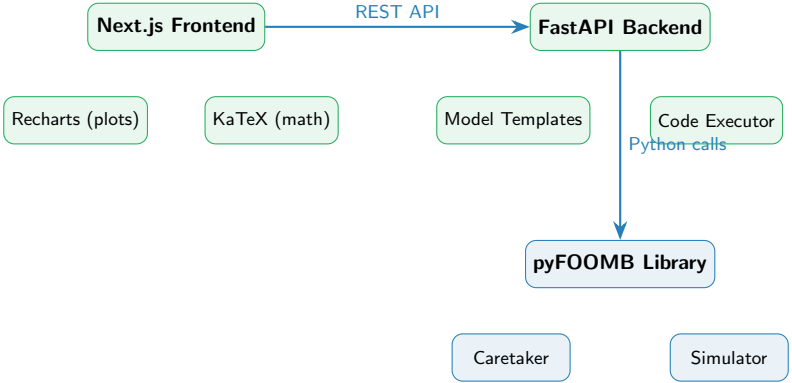
Our Solution: Full-stack web GUI wrapping the entire pyFOOMB API.

Layer	Tech
Frontend	Next.js, React, Recharts
Backend	FastAPI, Python
Solver	assimulo, pygmo, scipy

Features implemented:

1. 8 pre-built model templates
2. Interactive simulation with live plots
3. Data import (paste / upload / Google Sheets)
4. 5 parameter estimation methods
5. Sensitivity, FIM, covariance analysis
6. Multi-replicate parameter mapping
7. 39 automated API tests

Web Architecture



GUI — Six-Page Workflow

The GUI guides the user through the complete bioprocess modelling pipeline:

1. Model
Pick a template
Set parameters

→

2. Simulate
Run forward ODE
View plots

→

3. Data
Import experimental measurements

→

4. Estimate
Fit model params to data

→

5. Analyse
Sensitivity, FIM
Intervals

→

6. Replicates
Share params across experiments

5 Estimation Methods:

- Local optimisation (scipy)
- Parallel global search (pygmo)
- Repeated local
- Monte Carlo parameter sampling
- Parallelised Monte Carlo

Analysis Outputs:

- Sensitivity of each state
- Fisher Information Matrix
- Covariance & correlation
- Parameter confidence intervals
- Optimal Experimental Design

GUI — Screenshot

[Place your GUI screenshot here]

Save as `gui_screenshot.png` in the same directory and uncomment the includegraphics line above.

μ–q_p Relationship Models

The μ–q_p relationship governs how fast a cell produces product at a given growth rate. It determines the **optimal feeding strategy** in fed-batch.

Linear

$$q_p = \alpha\mu + \beta$$

Growth-associated.
Feed to maximise μ.

Bell-Shaped

$$q_p = q_p^{max} e^{-\frac{(\mu-\mu_{opt})^2}{2\sigma^2}}$$

Peak at μ_{opt}.
Maintain $\mu = \mu_{opt}$.

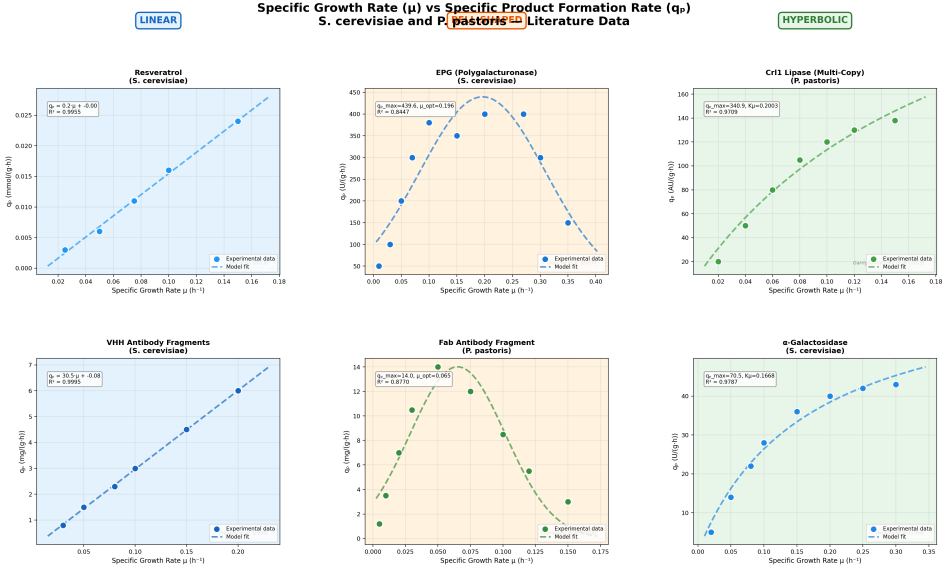
Hyperbolic

$$q_p = \frac{q_p^{max}\mu}{K_q+\mu}$$

Saturates at high μ.
Flexible feeding.

We surveyed 9 products from literature across *S. cerevisiae* and *P. pastoris*, and fitted each to one of these three models using `scipy.optimize.curve_fit`.

μ–q_p Fitting Results



Literature Survey — 9 Products, 2 Organisms

Data extracted from chemostat steady-state experiments.

Product	Organism	Model	Reference
Resveratrol	*S. cerevisiae*	Linear	Vos et al. (2015)
Heterologous Protein	*S. cerevisiae*	Linear	Liu et al. (2013)
Crl1 Lipase (SCC, PGAP)	*P. pastoris*	Linear	Nieto-Taype et al. (2020)
EPG Polygalacturonase	*S. cerevisiae*	Bell	Glauche et al. (2017)
Fab 3H6 (PGAP)	*P. pastoris*	Bell	Maurer / Garcia-Ortega
ROL Lipase (PAOX1)	*P. pastoris*	Bell	Garcia-Ortega et al. (2016)
Crl1 Lipase (MCC, PGAP)	*P. pastoris*	Hyperbolic	Nieto-Taype et al. (2020)
α-Galactosidase	*S. cerevisiae*	Hyperbolic	Giuseppin et al. (1993)
HSA	*P. pastoris*	Hyperbolic	Rebnegger et al. (2014)

Batch X, S, P Profile Simulation

From the fitted μ–q_p parameters, we simulated full batch dynamics for each product.

ODE System (Monod batch):

$$\frac{dX}{dt} = \mu \cdot X$$

$$\frac{dS}{dt} = -\frac{\mu}{Y_{X/S}} \cdot X$$

$$\frac{dP}{dt} = q_p(\mu) \cdot X$$

where $\mu = \mu_{max} \cdot \dfrac{S}{K_S + S}$

Integrated using `scipy.integrate.solve_ivp` (RK45).

Organism-Specific Parameters:

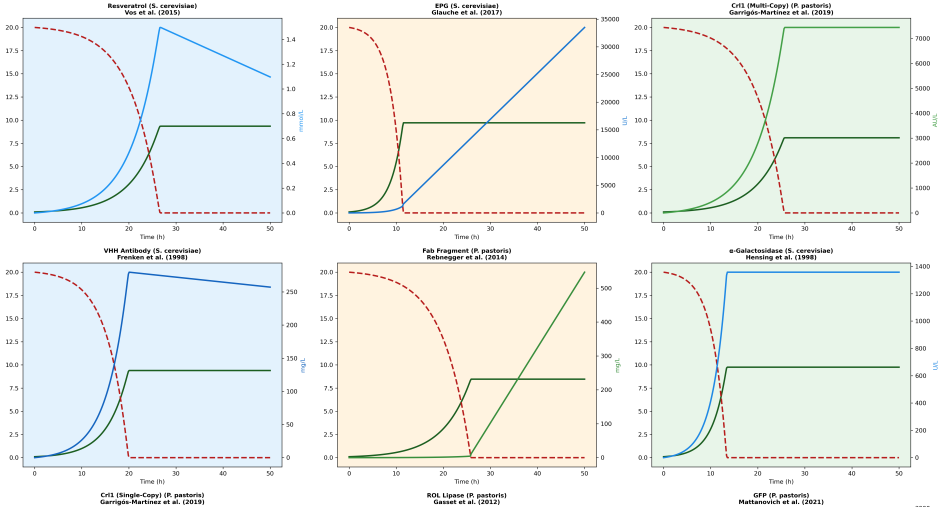
	S.c.	*P.p.*
μ_{max} (h^{-1})	0.40	0.20
K_S (g/L)	0.10	0.05
$Y_{X/S}$ (g/g)	0.50	0.45

$X_0 = 0.1$ g/L, $S_0 = 20$ g/L, $P_0 = 0$

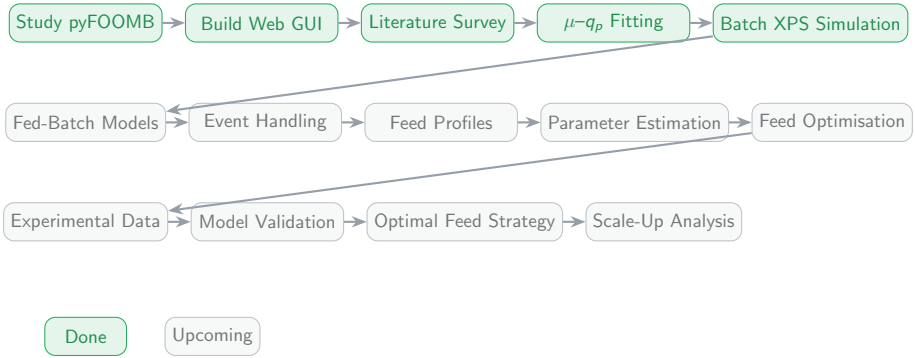
Note: While q_p data came from CSTRs (where $D = \mu$), the resulting $q_p(\mu)$ function is universal and

XPS Simulation Results

Validated XPS Profiles: Batch Simulations from Chemostat Kinetics



Project Roadmap



Thank You

Questions?

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